

CHAPTER 4

Activities & Lab Packet

Weather & Climate

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 4. Works with both the Standard and Easy Read versions of the textbook.

 Georgia Standard S6E4

WHAT'S INSIDE

- | | | |
|----|--|------------|
| 1 | Land vs. Water — The Uneven Heating Race | LAB |
| 2 | Convection in a Cup — Watching Warm Air Rise | LAB |
| 3 | Cloud in a Jar | LAB |
| 4 | Weather or Climate? | WORKSHEET |
| 5 | Layers of the Atmosphere | WORKSHEET |
| 6 | Front & Pressure Detective | WORKSHEET |
| 7 | Writing Prompts (3 levels) | WRITING |
| 8 | Classroom Discussion Questions | DISCUSSION |
| 9 | Be a TV Meteorologist! | GROUP |
| 10 | Vocabulary Bingo | REVIEW |

1 Land vs. Water — The Uneven Heating Race

LAB

OBJECTIVE

Measure and compare how fast land and water heat up and cool down — the root cause of wind and weather.

TIME

45–50 minutes

STANDARDS

S6E4.a

Materials (per group of 3–4)

- 2 identical cups: one filled with dry soil or sand ("land"), one with room-temperature water ("ocean") — same amount in each
- 2 thermometers, 1 desk lamp (the Sun), a ruler, a timer
- Student data sheet (next page)

Setup

A fair test is everything: same size cups, same fill level, same distance from the lamp (use the ruler — about 15 cm), thermometer bulbs at the same depth (just under the surface). The lamp must hit both cups equally.

Student Instructions

1. Record the starting temperature of both cups (minute 0).
2. Turn on the lamp so it shines equally on both cups. Record both temperatures every 2 minutes for 10 minutes ("heating phase").
3. Turn the lamp OFF. Keep recording both temperatures every 2 minutes for 10 more minutes ("cooling phase").
4. Graph both lines on the data sheet grid (sand = red, water = blue).
5. Answer the conclusion questions: which heated faster? Which cooled faster? Which held its temperature?

SAFETY & TEACHER NOTES

Safety: Lamp bulbs get hot — hands off, and keep cords taped down.

Expected result: Sand heats noticeably faster AND cools faster; water barely budes. If results are weak, move lamps closer (equally!) or use dark soil.

The payoff: This is the engine of the sea breeze and, scaled up, all global wind. Ask: "At the beach at noon, which is hotter — sand or sea? Now you have DATA."

ANSWER KEY (data sheet)

1) Sand/soil heated faster (bigger rise in 10 min) | 2) Sand also cooled faster; water changed slowly both ways | 3) Land = sand cup, ocean = water cup; at noon the land is hotter, at night the water is warmer | 4) Warm air over the hot land rises; cooler air over the water flows in to replace it = sea breeze | Conclusion: the Sun heats surfaces unevenly; that temperature difference sets air moving — the start of all wind and weather.

The Uneven Heating Race — Data Sheet

Name: _____

Period: _____

Time (min)	0	2	4	6	8	10 (lamp OFF)	12	14	16	18	20
SAND °C											
WATER °C											

Graph both lines here (label axes: time →, temperature ↑; sand = red, water = blue)

1. Which cup heated up faster while the lamp was on? By how many degrees did each rise?

2. Which cup cooled faster after the lamp went off?

3. At a real beach: which is hotter at noon — the sand or the sea? Which is warmer at midnight? Use your data.

Conclusion: Use your results to explain how a daytime sea breeze forms. (Words to use: rises, cooler, flows in.)

2 Convection in a Cup — Watching Warm Air Rise

LAB

OBJECTIVE

Watch warm fluid rise and cool fluid sink — the motion that creates wind and drives weather.

TIME

30 minutes

STANDARDS

S6E4.a, S6E4.b

Materials (per group of 3–4)

- 1 large clear container of **cold** water (add a few ice cubes 5 min early, then remove)
- 1 small bottle or film canister of **hot** tap water dyed deep **red**
- (Optional second round) a small bottle of ice-cold water dyed **blue**
- String or tongs for lowering the bottle; student data sheet (next page)

Setup

Hot tap water only — no kettles needed. The red bottle gets lowered gently to the **BOTTOM** of the cold tank, then uncapped (or opened with a thumb-release). Practice the lowering move with plain water first; smooth and slow wins.

Student Instructions

1. **Predict:** when the hot red water is released at the bottom of the cold tank, what will it do? Write your prediction first.
2. Gently lower the capped red bottle to the bottom. Let the tank go still.
3. Uncap/release and observe without bumping the table. Sketch what the red water does in the first 30 seconds, then after 2 minutes.
4. (Round 2) Float the blue ice-cold bottle at the **TOP** and release. Sketch what the blue water does.
5. Connect: red = warm air, blue = cool air, tank = the atmosphere. Complete the data sheet.

SAFETY & TEACHER NOTES

Safety: Hot tap water can still surprise — handle bottles with care; wipe spills fast.

Expected result: Red plume rises and spreads at the surface; blue sinks and spreads at the bottom. Together they trace a convection loop.

The vocabulary move: When the red rises, have students chorus "warm air **RISES** — LOW pressure." When blue sinks: "cool air **SINKS** — HIGH pressure." Wind = air flowing between the two.

ANSWER KEY (data sheet)

1) The red (hot) water rose to the top and spread out | 2) The blue (cold) water sank and spread along the bottom | 3) Warm = lighter/less dense so it rises; cool = heavier/denser so it sinks | 4) Rising warm air = low pressure; sinking cool air = high pressure; air flowing from high to low = wind | Conclusion: uneven heating makes some air warm (rises) and some cool (sinks in to replace it) — that circulating motion is what we feel as wind.

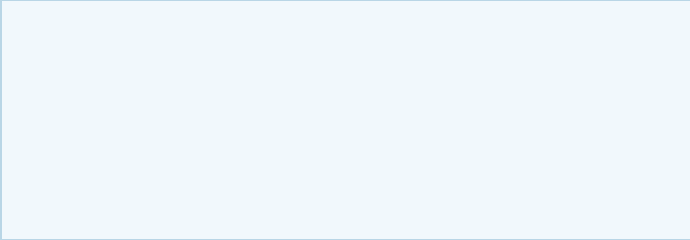
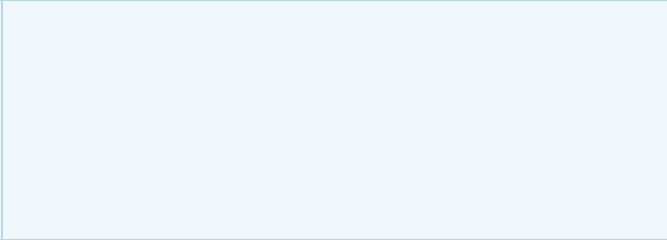
Convection in a Cup — Data Sheet

Name: _____

Period: _____

My prediction: When the hot red water is released at the bottom, it will...

Observations — sketch and label:

RED (hot) — first 30 seconds	RED (hot) — after 2 minutes
	
BLUE (cold) — first 30 seconds	BLUE (cold) — after 2 minutes
	

1. What did the HOT red water do when released? Was your prediction right?

2. What did the COLD blue water do?

3. Why does warm fluid rise while cold fluid sinks? (Hint: think heavy vs. light.)

4. Fill in: rising warm air creates _____ pressure; sinking cool air creates _____ pressure; air flowing from _____ to _____ pressure is what we call _____.

Conclusion: How does this tank explain why the wind blows on Earth?

3 Cloud in a Jar

LAB

OBJECTIVE

Create a real cloud and identify the ingredients every storm needs: moisture, cooling, and rising air.

TIME

25–30 minutes

STANDARDS

S6E4.c, S6E4.d

Materials (per group, or as a teacher demo)

- 1 clear glass jar with a metal lid (or a plate that seals the top)
- Warm water (about 2–3 cm in the bottom of the jar)
- Ice cubes (to pile on the upside-down lid/plate)
- Aerosol hairspray (teacher-controlled — one short puff)
- Dark paper backdrop; student data sheet (next page)

Setup

This works best as a teacher-led demo at each table, with students running the observations and the data sheet. The hairspray provides "condensation nuclei" — tiny particles for water vapor to grab onto, like dust in real clouds.

Student Instructions

1. Pour warm water into the jar and swirl. What's rising off the water? (Invisible water vapor — evaporation!)
2. Flip the lid upside down on top of the jar and pile ice on it. Predict: what will the cold "ceiling" do?
3. Teacher lifts the lid, gives ONE short puff of hairspray, and reseals fast.
4. Watch against the dark paper: a swirling cloud forms inside! Observe for 2–3 minutes, then lift the lid and watch the cloud escape.
5. On your data sheet, match each part of the jar to a weather ingredient: warm water = ?, ice lid = ?, hairspray = ?

⚠ SAFETY & TEACHER NOTES

Safety: Hairspray is flammable — no flames anywhere, teacher sprays, room ventilated. Warm (not boiling) water only.

The connection: Warm moist air rising + cooling aloft is EXACTLY how storm clouds build along a cold front (Section 4.5) and how hurricanes feed over warm oceans (4.6).

If no hairspray: a lit-then-blown-out match dropped in by the teacher also works (extra caution).

ANSWER KEY (data sheet)

1) Evaporation — warm water putting vapor into the air | 2) The cold lid cools the rising vapor = condensation | 3) Hairspray gives particles for droplets to form on (like dust in the sky) | 4) Warm water = warm ocean/ground moisture; ice lid = cold upper air; rising vapor = warm air rising | Conclusion: clouds need moisture + rising air + cooling; that's why storms form where warm moist air gets lifted fast (cold fronts, warm oceans).

Cloud in a Jar — Data Sheet

Name: _____

Period: _____

My prediction: When the ice-cold lid goes on top of the warm, moist jar...

Sketch the jar at each stage (label the cloud!):

Before hairspray	1 minute after	Lid lifted

1. What was the warm water doing to the air inside the jar? (Name the water-cycle step.)

2. What job did the icy lid do? (Name this step too.)

3. Why did the cloud only appear AFTER the hairspray puff? What does hairspray stand in for from the real sky?

4. Match the model to the real atmosphere:

In the jar	In the real sky
Warm water at the bottom	
Ice-cold lid up top	
Vapor rising through the jar	

Conclusion: A thunderstorm needs warm, moist air rising fast. Where in your jar did you see each of those three ingredients?

4 Weather or Climate?

WORKSHEET

OBJECTIVE

Distinguish day-to-day weather from long-term climate.

TIME

15 minutes

STANDARDS

S6E4

Name: _____

Period: _____

Part A. Write **W** (weather) or **C** (climate):

____ 1. It's raining in Atlanta right now

____ 5. Today's high will be 87°F

____ 2. Georgia summers are hot and humid

____ 6. The Arctic is cold all year long

____ 3. A cold front will arrive Thursday

____ 7. There's a tornado watch this afternoon

____ 4. Deserts get very little rain each year

____ 8. Hawaii stays warm in every season

Part B. Complete the table:

	WEATHER	CLIMATE
Time frame		
How fast it changes		
One example		

Part C. Short answers:

9. Explain the "mood vs. personality" memory trick in your own words.

10. A friend says: "It snowed in Georgia this week, so Georgia must have a cold climate." What's wrong with that thinking?

11. About how many years of data do scientists average to describe a climate?

ANSWER KEY

1) W 2) C 3) W 4) C 5) W 6) C 7) W 8) C | B: weather — right now/this week, changes hour to hour, e.g. "storming this afternoon"; climate — 30+ year average, changes slowly, e.g. "summers here are humid" | 9) Mood changes fast like weather; personality is the long-term pattern like climate | 10) One snowy week is weather — a single event can't change the long-term average pattern | 11) About 30 or more years.

WORKSHEET

A: troposphere 1, stratosphere 2, mesosphere 3, thermosphere 4, exosphere 5 | 6) troposphere 7) stratosphere 8) mesosphere 9) thermosphere 10) exosphere | 11) Nitrogen ~78%, oxygen ~21% | 12) The Sun's harmful UV rays; sunscreen for the planet | 13) They let sunlight in and trap some of the heat trying to escape | 14) No — with none, Earth would be a frozen ball; the problem is EXTRA gases trapping too much heat.

6 Front & Pressure Detective

WORKSHEET

OBJECTIVE

Use fronts and pressure to explain — and predict — the weather like a forecaster.

TIME

20 minutes

STANDARDS

S6E4.d

Name: _____

Period: _____

Part A. Write **CF** (cold front), **WF** (warm front), **H** (high pressure), or **L** (low pressure):

- | | |
|---|---|
| _____ 1. Cold air pushes under warm air | _____ 5. Cool sinking air; clear skies |
| _____ 2. Warm air slides slowly over cold air | _____ 6. Warm rising air; clouds and storms |
| _____ 3. Brings sudden, strong storms | _____ 7. After it passes, the air is cooler and clearer |
| _____ 4. Brings long, light, steady rain | _____ 8. The "calm, sunny weekend" pressure |

Part B. Forecast time! Read each report and predict the weather:

9. "A cold front will cross Georgia tonight." What should people expect tonight, and what will tomorrow feel like?

10. "A warm front is slowly approaching from the south." What kind of rain is coming, and for how long?

11. "The barometer shows high pressure building for the weekend." Picnic or board games inside? Why?

Part C. 12. A thunderstorm needs two main ingredients. Name them — and explain what EXTRA thing must happen for a tornado to form.

13. What is an air mass, and what do we call the boundary where two different ones meet?

ANSWER KEY

1) CF 2) WF 3) CF 4) WF 5) H 6) L 7) CF 8) H | 9) Sudden strong storms tonight; cooler, clearer air tomorrow | 10) Long periods of light, steady rain, then warmer air | 11) Picnic — high pressure = sinking cool air = clear skies | 12) Warm, moist air + rising fast; for a tornado the rising air must also start to SPIN | 13) A large body of air with similar temperature and moisture; a front.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain concepts in writing.

TIME

15–30 minutes each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Mood or Personality?** Explain the difference between weather and climate using the mood/personality trick — then give one example of each for the place where YOU live.
2. **Elevator Through the Sky.** You ride a magic elevator from the ground to outer space. Describe what you pass in each layer of the atmosphere, in order.
3. **Why the Wind Blows.** A kindergartner asks you, "Where does wind come from?" Explain it simply but correctly, using the idea that warm air rises.

Level 2 — Intermediate (3 paragraphs)

4. **The Sea Breeze Mystery.** Every sunny afternoon at the beach, a cool wind starts blowing in from the ocean. Explain exactly why, step by step: uneven heating → rising air → pressure → wind.
5. **Front Wars.** A cold front and a warm front each claim to be "the most dramatic." Write their argument — each must accurately describe how it moves, what weather it brings, and what comes after it.
6. **Anatomy of a Hurricane.** Explain a hurricane's life story: where it's born, what fuels it, why it strengthens over water, and why it fades over land. Include the word "eye."

Level 3 — Advanced (essay)

7. **The Chain Reaction.** Defend this claim with evidence from the whole chapter: "Every weather event on Earth ultimately starts with the Sun heating the planet unevenly." Trace the chain from sunlight all the way to a thunderstorm.
8. **Warmer Oceans, Stronger Storms?** Hurricanes are fueled by warm ocean water. Write an essay predicting what could happen to hurricanes if oceans keep warming — and explain the science behind your prediction.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific thinking through verbal reasoning.

USE

Warm-ups, exit tickets, Socratic seminar

STANDARDS

Cross-cutting

TEACHER NOTES

These are designed for "think-pair-share" format. Give students 30 seconds to think, 1 minute to discuss with a partner, then call on a few groups. Some questions have no single "right" answer — push students to defend their thinking with evidence from the chapter.

Quick Warm-Ups (1–2 min each)

1. Your friend says "climate change isn't real because it was cold yesterday." What word are they confusing?
2. Why do jets fly in the stratosphere instead of the troposphere?
3. Concrete sidewalks burn your feet in summer, but the pool doesn't. What chapter idea is that?
4. The weather report says "low pressure moving in." Do you grab sunglasses or an umbrella?
5. Why don't hurricanes ever form over Kansas?

Deeper Discussions (5–15 min each)

6. Greenhouse gases are described as both life-saving (Earth would freeze without them) and worrying (too much warms the planet). How can both be true at once?
7. Cities are often a few degrees hotter than the countryside around them. Using land vs. water heating ideas, brainstorm why — and what could cool a city down.
8. Georgia gets tornadoes mostly in spring. What is colliding in spring that makes the storms so strong? (Hint: think air masses.)
9. Forecasters can predict a hurricane days ahead, but a tornado only minutes ahead. What about each storm might explain that difference?
10. If Earth's surface were ALL ocean — no land at all — how would winds and weather be different?

9 Be a TV Meteorologist!

GROUP PROJECT

OBJECTIVE

Deliver a scientifically accurate TV-style weather forecast using fronts, pressure, and storm safety.

TIME

2 class periods + presentations

STANDARDS

S6E4.a, S6E4.d

Project Description

Teams of 3–4 become a TV weather team and present a 2–3 minute "broadcast" for a fictional week in Georgia. The teacher hands each team a secret "weather scenario card" (e.g., "a strong cold front arrives Wednesday" or "high pressure parks over the state all week" or "a weakening hurricane approaches the coast"). The team must turn the science into a forecast the public can use.

Broadcast Requirements

- A hand-drawn or digital **weather map** with front symbols, H and L pressure centers, and temperatures
- A day-by-day forecast that correctly matches the science of the scenario card
- Correct use of at least **5 vocabulary words** (front, air mass, pressure, humidity, etc.)
- One **safety tip** matched to the forecast (thunderstorm, tornado, or hurricane safety)
- Every team member speaks on camera (or "on air")

Presentation Day

Teams broadcast live or play a recorded video. The audience acts as "viewers at home" and asks one call-in question per broadcast. Class votes on awards: Most Accurate, Best Map, Best On-Air Energy.

TEACHER NOTES

Differentiation: Provide a script skeleton ("Today... Tomorrow... The reason is...") for teams who need structure; advanced teams must also explain WHY their front brings that weather.

Science check: Watch for cold fronts bringing "all-day drizzle" (that's warm front weather!) — catch it at the map-draft stage.

Grading rubric (suggested): Scientific accuracy (40%), required elements (25%), creativity (20%), presentation (15%).

10 Vocabulary Bingo

REVIEW GAME

OBJECTIVE

Review chapter vocabulary in a low-stress game format.

TIME

20–30 minutes

STANDARDS

S6E4 (all)

How to Play

1. Students write one word from the word bank into each square of their board, in any order. (FREE space stays free.)
2. The teacher reads a **definition** (not the word!). Students mark the square if they have the matching word.
3. First to 5 in a row calls "BINGO!" — but must read back each marked word AND its meaning to win.
4. Play again with "blackout" rules (whole board) for a longer review.

Word Bank (24 words + FREE)

weather	greenhouse gas	sea breeze	tornado
climate	greenhouse effect	air mass	hurricane
atmosphere	air pressure	front	humidity
troposphere	high pressure	cold front	mesosphere
stratosphere	low pressure	warm front	thermosphere
ozone layer	wind	thunderstorm	exosphere

		FREE 		

TEACHER NOTES

Read definitions straight from the Vocabulary Reference sheet (Section 4) so wording matches what students studied. Keep a checklist of called words to verify BINGOs. For struggling readers, display the word bank on the board during play.